Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/10/2005
PI Name:	Ceperley, David M. Ph.D.		
Project Title:	Properties of liquid helium from computer simulation		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	61801	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
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No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Ceperley, David		
COI Name (Institution):	Goldenfeld, Nigel (University of Illinois Urbana-Champaign)		
Grant/Contract No.:	NNC04GA01G		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		
Task Description:	We are conducting accurate simulations, based on realistic interatomic potentials, of bulk superfluid, superfluids in restricted geometeries and arrays of vortices. The aim of this research is to provide reliable, quantitative theoretical predictions which will aid in the design and interpretation of low temperature experiments performed in the microgravity facility. Considerable progress can be achieved on these fronts, owing to the possibility of performing accurate measurements in a gravity- controlled environment and making comparison with state-of-the-art numerical calculations. This research is concerned with fundamental aspects of helium and quantum fluids in general; we address outstanding problems in the current understanding of relevant phenomena such as Bose condensation, superfluidity, and phase transitions, as well as of theoretical issues such as the inference of bulk properties of matter from the study of finite clusters. The theoretical issues involved in helium systems are of direct relevance to understanding other many-body quantum systems such as correlated electronic systems		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	This project aims to study fast rotating superfluids. In particular it focuses on Bose Einstein Condensates (BEC) in alkali gases, which are so eagerly studied both theoretically and experimentally. These rotating superfluid gases form an array of vortices, like in the case of superconductors under a magnetic field. Unlike superconductors though, the destruction of the superfluid state is not due to the overlap of vortices and destruction of the bosons (cooper pairs). In fact, mean field calculations (Phys. Rev. Lett. 90, 140402 (2003)) suggest that the core size reduces in such a way that cores never overlap. Obviously, this description must break down for high rotation rates when core sizes reduce below the minimal length described by the mean field theory. To understand how this happens is the goal of this project. Path Integral Monte Carlo (PIMC) simulations involve no uncontrolled approximations and are therefore ideally suited for this task. We have adapted the PIMC code to deal with systems lacking time reversal symmetry (such as vortex arrays) and to obtain the vortex core structure. We have implemented finite difference methods to provide a consistency check for our simulations with the theoretical predictions of the main tool for studying Bose Einstein Condensates: the Gross-Pitaevskii equation. Results are in agreement with a modified version of this equation applicable to the densities we have studied. Interestingly, the unmodified version of this equation also seems to give the right vortex profile as long as a proper scaling is done. It therefore seems that the Gross-Pitaevskii equation has a wider range of validity than expected, or at least can be simply renormalized to extend its validity. The exact results from the PIMC calculations have been essential to realize this conclusion. The PIMC also has allowed us to go into a regime where the vortex profile is very different from what would be expected from the Gross-Pitaevskii equation. In ongoing work, which we plan to complete early in the forthcoming grant period, we would like to address the case of extreme low density, when vortex cores are about the size of the cell and check whether the Gross-Pitaevskii mean field description breaks down. We will also calculate the energies of the vortex array, in order to assess when the array would become unstable. Finally, we would like to examine the two dimensional condensate. The theoretical issue here is that there is not agreement on the derivation of the Gross-Pitaevskii equation in two dimensions, where the scattering problem becomes subtle. As a result the dependence of the quartic coupling constant on the scattering length is controversial. PIMC can possibly resolve the debate by calculating precise vortex profiles and comparing with numerical calculations based on the various forms of the Gross-Pitaevskii equation. Another project was motivated by the observation of a possible supersolid state of helium in porous Vycor (E. Kim and M. H. Chan, Nature 427, 225 (2004). The possibility of supersolid has been suggested more than 30 years ago but never observed experimentally. We simulate the system using Path_Integral_Monte_Carlo. Our model gives qualitative results agreeing with experiments on Helium-Vycor system. The Helium atoms tend to form high density layers near the Vycor walls and layers with a perfect hcp crystal structure inside the pores. In between, there are at least two glassy layers close to the Vycor surface. We predict those layers are the superfluid in this system. We are now studying the role of 3He in quenching this effect. We are at the end of this project and expect to submit our results for publication this year. We have also been examining the extent to which bulk 4He can be a supersolid and have ruled out mechanisms involving exchange in a perfect crystal and point defects and are examining the possibility of a dislocation mechanism.
Bibliography Type:	Description: (Last Updated: 09/11/2009)
Articles in Peer-reviewed Journals	Draeger EW, Ceperley DM. "Superfludity in a doped helium droplet." Physical Review Letters. 2003 Feb;90(6):065301. http://dx.doi.org/10.1103/PhysRevLett.90.065301 , Feb-2004